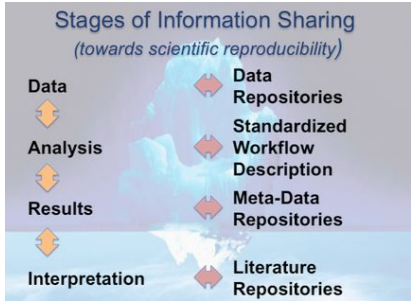
Big Data to Knowledge(BD2K)

Best Practices for Biomedical Research Data Management

Module 9

Scientific Research Team



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Learning Objectives

1. Describe a project where researchers and a librarian worked together to find a way to uniquely identify data
2. Understand how librarian skills used in the past to identify and catalog publications can be easily transferred to doing the same for data objects and collections of data objects
3. Recognize the potential of partnerships between librarians and researchers
4. Identify different roles librarians may play with research teams
5. Describe new professional skills librarians and researchers can develop through working together

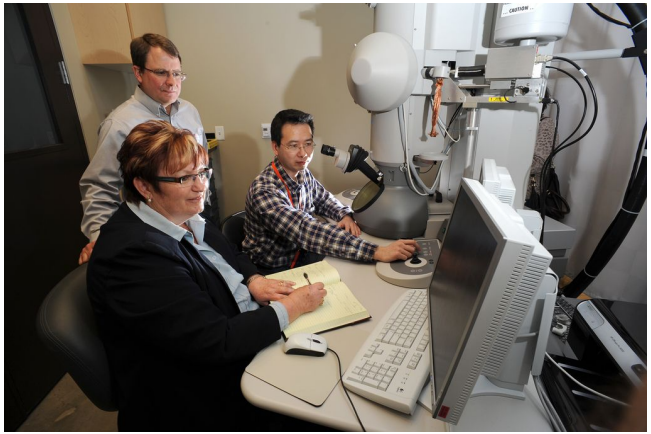
Structure of a Research Team

Diversity:

- Co-investigators, technical and clerical staff and postgraduate research students
- Disciplinary backgrounds, motivations and aspirations, and cultural backgrounds

Balance:

- Skills, expertise and other contributions appropriate to achieve the team's objectives
- Behavioural traits or work characteristics aligned to particular



Team Science

Team science is a collaborative effort to address a scientific challenge that leverages the strengths and expertise of professionals trained in different fields. Coordinated teams of investigators with diverse skills and knowledge may be especially helpful for complex studies.

Collaborative Scientific Research Project

The National Library of Medicine awards supplemental grants to NIH grantees to include an informationist (academic librarian) to contribute library science skills to the research project. The purposes of this program are:

- to enhance collaborative, multi-disciplinary basic and clinical research by integrating an information specialist into the research team in order to improve the capture, storage, organization, management, integration, presentation and dissemination of biomedical research data
- to assess and document the value and impact of the informationist's participation

Positive Feedback

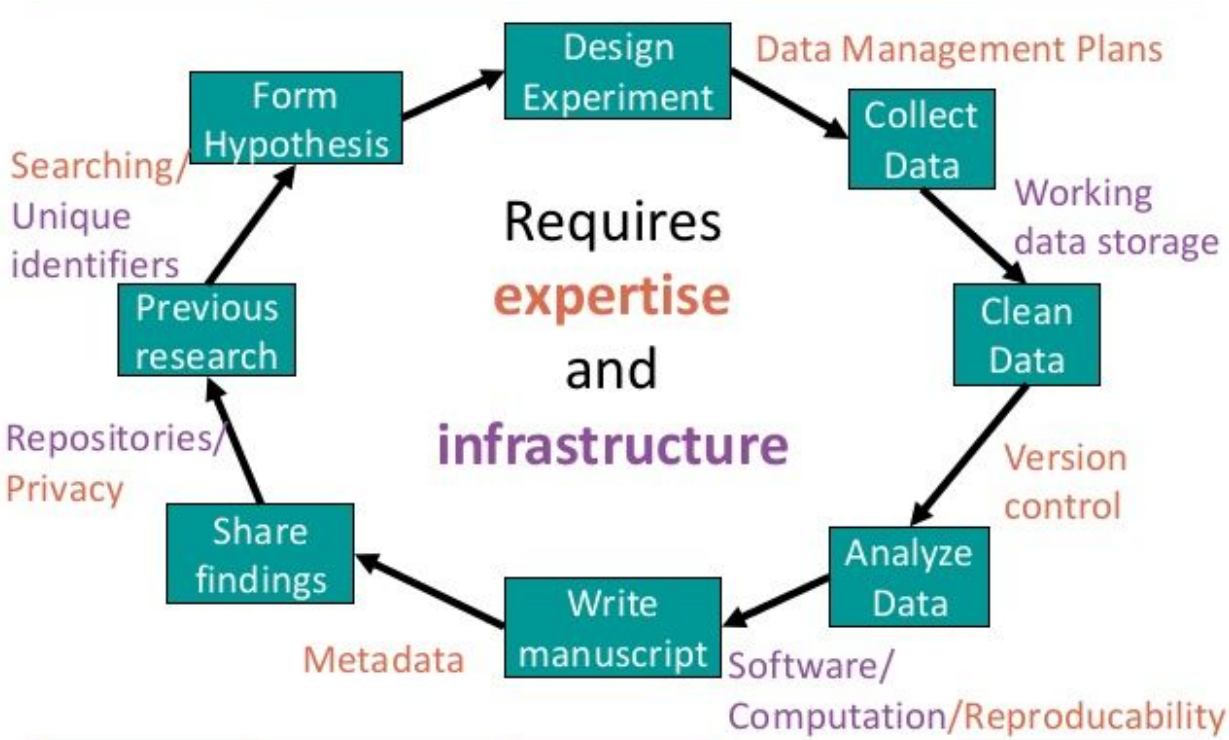
"The main goal of the NLM administrative supplement award is to enhance biomedical research through the integration of an informationist focused on data management. The results of the survey and focus group show that informationists can make many contributions to this area. **According to the PIs, informationists had a positive impact on their research process and helped them make progress toward their research goals. When considering the data management issues that were the focus of the award, informationists had impact in several areas, including data storage, data archiving, and data management planning, and were able to help the PIs make informed decisions about their data.** Moreover, the informationists were able to contribute their expertise to other areas such as literature searching, which helped the team make progress toward their research outcomes."

"Assessing the National Library of Medicine's Informationist Awards" http://dx.doi.org/10.7191/jeslib.2016.1095

How a Librarian/Informationist can Assist in Research Projects

- Literature searching
- Citation assistance
- Database design
- Data management
- Connect with other useful services on campus
- Co-authoring publications

Roles within the Lifecycle



CU Anschutz Health Science Library Data Services
http://www.slideshare.net/CTobinMagle/cu-anschutz-health-science-library-data-services

Example Project

The videos in this module provide both a researcher and a librarian perspective from collaborating on a project to:

- Develop a standard for how to uniquely identify and cite shared/reused data so the data creator is credited
- Use a neuroimaging repository developed by the researcher as a case study
- Project located at University of Massachusetts Medical School Department of Psychiatry